Columbia Local Services Database - Project Report

CS 284 / CS 430 Project 1 | Fall 2026 | Team: Ocean Panta, Samir Khanal, Jonathan Greer

1. Scope, Business Rules, and Data Sources

User and scope. The database serves residents and newcomers in Columbia, MO (including international students) who want to find a provider for oil/inspection and exhaust service or for imported and specialty groceries, compare services *within our collected sample*, and trace any offering back to its source. We include one physical location per business, a sourced name and address, and an offering only when a source supports it. Categories: Automotive and Specialty Markets.

Sources. Raw observations are in data_collection/raw_data.csv (15 rows, each with a raw-row ID, URL, date and collector). Auto shops come from each shop's own website; the seven markets come from the Mizzou International Student Services (ISSS) food-store page. Directories (Loop Como, FindGlocal, Chamber of Commerce) were used for discovery and for the three address cross-checks documented in corraboration.txt. All 14 registered sources are in sources.csv. Cleaning rules, before/after examples and counts are in rules.txt: 15 collected, 15 kept, none excluded.

Four business rules.

1. **Identity:** each Business row is one physical location (business_id); business_name is UNIQUE, and each business has one source_id and the raw_row_id where it was observed.

2. **Relationships:** a Service belongs to exactly one Service_Category (one-to-many); businesses and services are many-to-many through Business_Service, which also carries the source_id and raw_row_id for each offering.

3. **Required values:** business_name, street_address, city, state, source_id and raw_row_id are NOT NULL; website and postal_code are optional and are NULL when not verified.

4. **Duplicate prevention:** Business_Service has composite primary key (business_id, service_id); category_name, service_name and source URL are UNIQUE. Foreign keys use the MySQL default RESTRICT, so a parent row cannot be deleted while children reference it.

Item	Collected	Final	Minimum
Businesses	15	15	15
Service types	4	4	4
Categories	2	2	2
Business-service pairs	20	20	20
Distinct publishers (names)	14	14	2
Corroborated businesses	3	3	3

Table 1: Dataset counts. Publisher names are distinct; the auto-shop sites and ISSS are different organizations.

2. Model and Key Schema Choice

Five tables: Business, Service_Category, Service, Business_Service (associative) and Source; the ER diagram is on page 3 and the README lists each table's purpose, primary key and foreign keys. **Key choice:** Source is its own table referenced by both Business and Business_Service. A business's identity/location and each offering can therefore cite different sources, and the source URL, publisher and date are stored once. Because evidence is business-level, results mean "business offering recorded; branch availability unverified."

3. Normalization Example (UNF to 3NF)

We use raw observation 3 (Midas Muffler Shop). **Combined relation:** R(raw_row_id, business_name, street_address, city, state, postal_code, service_name, category_name, source_url, publisher, title, access_date, raw_file). **Candidate key:** (raw_row_id, service_name). **Dependencies:** raw_row_id -> business_name, street_address, city, state, postal_code, source_url; service_name -> category_name; source_url -> publisher, title, access_date, raw_file.

Decomposition. 1NF: one service per row and address split into parts. 2NF: business facts depend only on raw_row_id and category only on service_name, so they move to Business and Service. 3NF: source_url -> publisher details is a transitive dependency, so Source becomes its own table and Business keeps source_id; category_name is stored once in Service_Category.

Anomaly removed. In the combined table Midas has two rows (oil/inspection and muffler/exhaust), so its address and source details are repeated; fixing the address in one row but not the other would give two addresses for one shop (update anomaly). In the final schema the address is stored once in Business. **Reconnection:** Business_Service joins Business on business_id, Service on service_id (then Service_Category on category_id) and Source on source_id.

4. Findings, Limitations, and Follow-up

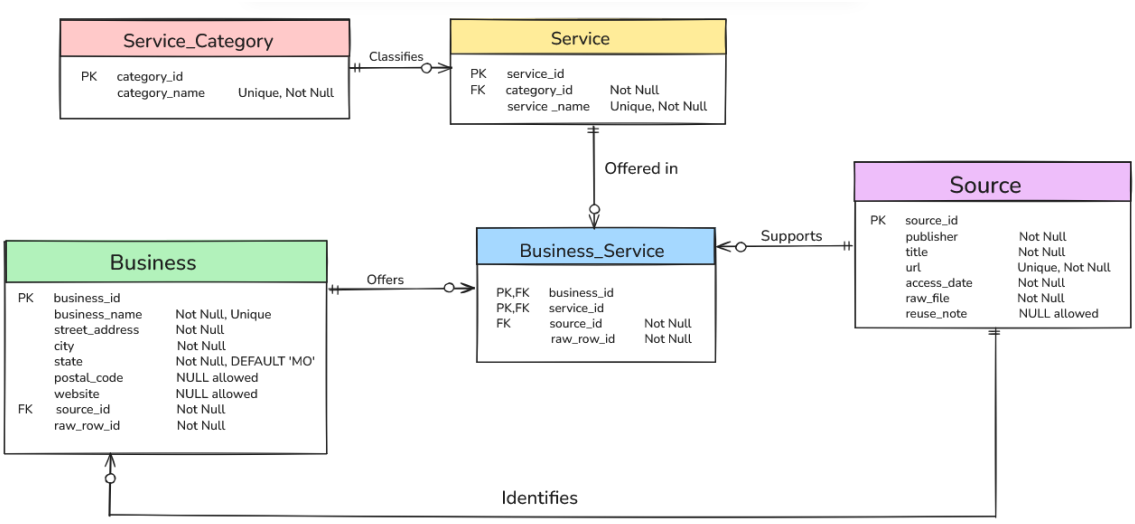
Finding 1 (Q3, threshold 3). Three services have at least three recorded providers: oil/filter change and inspection (8, all eight auto shops), imported packaged groceries (7, all seven markets) and muffler/exhaust (3 of 8 shops). In-store fresh food/deli has only 2 (A&Y Global Market and Tienda El Patron), so it is excluded. Q2 lists the 8 Columbia providers for oil/inspection, each with its source URL and raw-row ID. A provider count in a small sample does not show unmet market demand.

Finding 2 (Q6). Six of 15 businesses (40%) have no recorded website: African Market, Go-Fresco, Iyumva, Sai Market, Shwe Market and Tienda El Patron. All six are specialty markets, and they are documented only through the ISSS page. Q4 returns the seven markets with no recorded oil-change offering, which is expected: "not recorded" does not mean the business lacks the service.

Limitations. The sample is small and non-random: markets come from one university-oriented list and auto shops from sites we happened to find. Evidence is business-level and each offering rests on one source, with the three cross-checks covering addresses only. Missing website means none found, not none exists. **Follow-up question (hypothesis, not tested):** do the six markets without a website have another online presence, such as a social page, that would make them easier to verify?

Validation. T1 confirmed the counts and source links on the real snapshot; T2 showed error 1452 for an offering with a nonexistent business; T3 showed error 1062 for a duplicate pair; T4 used a self-cleaning fixture for Q3-Q6 behavior. Outputs: query_output.pdf; log: validation_logs.txt. Contributions and the AI-use statement are in the README.

Entity-Relationship Diagram (crow's foot notation)



This diagram shows the ER diagram using crow's foot notation for Columbia Local Services Database
Double bar means compulsory exactly one parent,
Crow's foot with circle means zero or many children
In our case it means, Business to Business Service - 1:N, Business to Source - N:1 with identifies as a relation in between

Service_Category 1:N Service; Service 1:N Business_Service; Business 1:N Business_Service; Source 1:N Business; Source 1:N Business_Service. Business_Service has composite primary key (business_id, service_id) and all foreign keys are mandatory. Website and postal_code are optional.